Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2024

Course: B.Tech | *Time: 03 Hours*
Code: 155503 | *Subject: Artificial Intelligence* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Which of the following best defines the Turing Test?
- (i) A method to calculate machine efficiency
- (ii) A test for machine learning algorithms
- (iii) A test to determine if a machine can exhibit human-like intelligence
- (iv) A benchmark for robotic speed

(b) *The A** algorithm guarantees optimality if the heuristic used is
- (i) Arbitrary
- (ii) Inconsistent
- (iii) Admissible and consistent
- (iv) Random

(c) Which of these search strategies is *not* complete in infinite-depth spaces?
- (i) Breadth-First Search
- (ii) Depth-First Search
- (iii) DFID
- (iv) A* Search

(d) In an AND/OR graph, solving an AND node means:
- (i) Solving one child node is enough
- (ii) All child nodes must be solved
- (iii) The node is ignored
- (iv) Any node can be skipped

(e) Minimax is used in game playing to:
- (i) Maximize randomness
- (ii) Minimize evaluation time
- (iii) Make optimal moves assuming the opponent plays optimally
- (iv) Store all possible states

(f) Alpha-beta pruning improves Minimax by:
- (i) Increasing the depth
- (ii) Ignoring non-optimal branches
- (iii) Doubling the search time
- (iv) Reducing the score of nodes

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(g) Propositional logic cannot express which of the following?
- (i) True/false statements
- (ii) Compound statements
- (iii) Logical connectives
- (iv) Variables and functions

(h) Bayesian networks are especially useful in situations with:
- (i) Complete knowledge and logic
- (ii) No uncertainty
- (iii) Deterministic processes
- (iv) Probabilistic and uncertain information

(i) Which of the following is most likely used in deep learning?
- (i) Decision Trees
- (ii) Shallow neural networks
- (iii) Multi-layered neural networks
- (iv) K-Means algorithm

(j) Which of the following is a key step in a genetic algorithm?
- (i) Regression
- (ii) Cross-validation
- (iii) Crossover
- (iv) Tokenization

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Q.2 (a) A chatbot consistently passes the Turing Test during short conversations but fails at longer ones. What does this imply about the limitations of the test? Suggest an enhancement to the test. [7]

(b) Explain how an agent's architecture affects its ability to operate in a partially observable, stochastic environment. Provide a practical example. [7]

Q.3 (a) Show that Hill Climbing can get stuck in local maxima, plateaus, and ridges. Suggest at least two variations or techniques to overcome these issues and explain how they help. [7]

(b) Heuristics are designed to guide search algorithms efficiently, but they can sometimes mislead the search. Describe a situation in which A* search performs worse than an uninformed search due to a poorly chosen heuristic. [7]

Q.4 (a) Define a basic Constraint Satisfaction Problem (CSP) involving three variables and solve it using backtracking. Also explain the role of domain and constraints. [7]

(b) Describe how Particle Swarm Optimization (PSO) works with a simple analogy (e.g., birds flocking). Show how particles update their positions and velocities. [7]

Q.5 (a) Create a simple two-player game tree (3 levels) and show how the Minimax algorithm selects the best move for the maximizing player. [7]

(b) Compare probabilistic reasoning using Bayesian Networks with logical reasoning. In what situations is probabilistic reasoning more appropriate? Give examples. [7]

Q.6 (a) What is partial-order planning? Explain with a block-world example, showing how actions can be arranged without a strict linear order. [7]

(b) Explain resolution in First-Order Logic (FOL). Using a simple knowledge base (e.g., "All humans are mortal, Socrates is a human"), derive a conclusion using resolution refutation. [7]

Q.7 (a) Construct a simple decision tree from the given dataset: [7]

Weather	Temp	Play?
Sunny	Hot	No
Sunny	Cool	Yes
Rainy	Cool	Yes
Rainy	Hot	No

Explain your attribute selection criteria.

(b) How can semi-supervised learning be useful in real-world scenarios like email spam detection? Describe its working with an example involving labeled and unlabeled data. [7]

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Q.8 (a) Discuss how backpropagation helps train a neural network. Use a small example with two layers and a loss function to explain weight adjustment. [7]

(b) What are the limitations of K-means clustering? Discuss two scenarios where K-means performs poorly and suggest possible improvements or alternatives. [7]

Q.9 (a) What is a Genetic Algorithm (GA)? Outline the basic steps of a GA such as selection, crossover, and mutation using an example of solving a simple optimization problem. [7]

(b) Explain the working of rule-based Expert Systems. Create a small rule base for diagnosing common cold vs. flu using IF-THEN rules. [7]

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